

Dang Nguyen

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Research Interests

- **AI alignment:** understanding the behavior and values of LLMs and how to align them with human values.
- **LLM communication:** improving LLMs’ ability to communicate and act in high-stakes settings.
- **LLM creativity:** understanding and improving LLMs’ ability to create novel content, ideas, ways of acting.
- **Human-centered AI:** understanding human-AI interaction and how to design AI that improves humanity.

Education

2024 - Exp. 2029	Computer Science Ph.D. , The University of Chicago. Advisor: Prof. Chenhao Tan
2020 - 2024	Computer Science B.S. & Mathematics B.A. , The University of Chicago. Specialization in Machine Learning.

Selected Publications

2025	On the Effectiveness and Generalization of Race Representations for Debiasing High-Stakes Decisions. [Link] Nguyen, D. , & Tan, C. In Conference on Language Modeling (COLM). <i>Top 1% of papers in the review process. Best Poster Award runner-up at Midwest Machine Learning Symposium 2025.</i> Keywords: fairness and bias, mechanistic interpretability, causal interventions
2025	GPT-4V Cannot Generate Radiology Reports Yet. [Link] Jiang, Y.*, Chen, C.*, Nguyen, D. , Mervak, B., & Tan, C. In Findings of the Association for Computational Linguistics (NAACL). <i>Lightning talk at the Toyota Technological Institute at Chicago (TTIC) Multimodal AI Workshop 2024.</i> Keywords: evaluation of large vision-language models, prompt engineering, error analysis
2023	Pragmatic Radiology Report Generation. [Link] Nguyen, D. , Chen, C., He, H., & Tan, C. In Proceedings of Machine Learning for Health (ML4H). <i>Example paper for ML4H 2024.</i> Keywords: vision-language modeling, finetuning LLMs, reducing hallucinations

Research Experience

2025 - Present	Communication and Intelligence , Graduate Researcher. Advisor: Prof. Ari Holtzman, Prof. Chenhao Tan Project title: Benchmarking AI communication with Communication Games. • Developed HR Simulator, an email-based game to benchmark human and AI communication. Gained 200+ players. Hosted at hrsimulator.communicationgames.ai. • Analyzed 600+ human and LLM-written emails across 5 writer models and 20 LLM judges. • Found LLM email readers consistently prefer LLM emails over human emails, but human+LLM emails outperform both. • Found evidence for “emergent tact”: smaller models prefer less tactful while larger ones prefer more tactful emails. • Used a Likert scale and human annotations to show LLMs predominantly write formal and empathetic emails while people write more tone-varied emails.
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2024 - 2025	Chicago Human+AI Lab, Graduate Researcher. Advisor: Prof. Chenhao Tan Project title: On the Effectiveness and Generalization of Race Representations for Debiasing High-Stakes Decisions. - Designed datasets to evaluate racial biases in Gemma and LLaMA 3.2’s decisions in college admissions and hiring. - Used Distributed Alignment Search to find models’ causal race representations with up to 90% accuracy. - Reduced models’ biases by up to 57% using representation-based interventions. - Showed that race representations can be prompt-specific, highlighting the challenge of limited generalizability.
2023 - 2024	Chicago Human+AI Lab, Undergraduate Researcher. Advisor: Prof. Chenhao Tan Project title: Pragmatic Radiology Report Generation. - Studied automated chest X-ray report generation using multi-modal LLMs. - Used a Chi-squared test to show that the indication section of a report drives mentions of negative findings. - Prompted Flan-T5 to clean reports, which reduces hallucinations by 69.5% from the previous state-of-the-art (SoTA). - Developed a vision-language model using ResNet and LLaMA 7B that outperforms the previous SoTA by 29% on disease identification, 43% in writing style, and 52.9% on negative findings reporting.
2020 - 2022	Machine Learning for Systems Research (UCARE Lab), Undergraduate Researcher. Advisor: Prof. Haryadi Gunawi - Developed statistical models to detect and analyze anomalous HDDs. - Used Pandas to process disk performance data and optimized the runtime to parse 1 billion data points by 90%. - Applied K-means and DBSCAN on 23 pairwise correlations of data features to identify slow disk clusters. - Developed a KL divergence-based algorithm to detect slow disks. - Analyzed 50,000 disks in 3 days using 3 high-performance compute servers.

Professional Experience

Summer 2022	NetApp Inc., Research Intern. - Developed machine learning algorithms to detect slow disk drives. - Optimized an algorithm based on KL divergence and improved NetApp’s threshold-based detector’s true positive rate from 10% to 59%. - Used Apache Hive and SQL to create 3 novel datasets of slow disks.
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Teaching Experience

Winter 2026	Teaching Assistant, CMSC 25700: Natural Language Processing, University of Chicago. - Designed homework assignments on Zipf’s Law and LLM Best-of-N Sampling. - Mentored 9 student teams on their final project over the course of 5 weeks.
Winter 2025	Teaching Assistant, CMSC 25700: Natural Language Processing, University of Chicago. - Designed homework assignments on transformers, pre-training, and interpretability. - Led a tutorial session on PyTorch. - Mentored 7 student teams on their final project over the course of 5 weeks. 3 advised teams ranked in the top 6 out of 26 teams, and 1 team won the best final report award.

Service

2025	Reviewer , Conference on Language Modeling (COLM).
2025	Reviewer , International Conference on Learning Representations (ICLR).
2024	Reviewer , Machine Learning for Health (ML4H).